

Fixed Point and Convergence Theorem of SCX Self-Evolution

Version: 2026-06-28 | **Status:** Theoretical framework (partial proof) | **Scope:** Central theorem establishing convergence regimes for the self-evolution loop **Notation note:** $\mathcal{S}$ denotes the state space (existing SCX theory). S_t denotes the gatekeeper scoring function at time t . $\mathcal{M}_t$ denotes the memory bank. f_{θ_t} denotes the NEP student.

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1. Introduction: The Central Question

The self-evolution loop couples three components:

$$\begin{aligned}
 \mathcal{M}_t &\leftarrow \text{AcceptOrReject}(\mathcal{D}_t, S_t) && \text{(gatekeeping)} \\
 \theta_{t+1} &\leftarrow \theta_t - \alpha_t \nabla_{\theta} \ell(f_{\theta_t}, \mathcal{M}_t) && \text{(student update)} \\
 S_{t+1} &\leftarrow \text{UpdateSCX}(S_t, \mathcal{M}_t, f_{\theta_{t+1}}) && \text{(gatekeeper update)}
 \end{aligned}$$

Central question: Does the sequence $(S_t, \theta_t, \mathcal{M}_t)$ converge? If so, to what? If not, what alternative long-term behaviors are possible?

This document provides the central theorem (SE-1) that characterizes the convergence properties of the self-evolution system.

2. Theorem SE-1: Convergence of SCX Self-Evolution (Formal Statement)

Theorem SE-1 (Convergence of SCX Self-Evolution). Let $(S_t, \theta_t, \mathcal{M}_t)$ be the sequence generated by the SCX self-evolution algorithm:

$$\begin{aligned}
 \mathcal{M}_{t+1} &= \mathcal{M}_t \cup \{(x, y) \in \mathcal{D}_{t+1} : S_t(x, y) \geq \gamma\} \\
 \theta_{t+1} &= \theta_t - \alpha_t \nabla_{\theta} \ell(f_{\theta_t}(x_t), y_t), \quad (x_t, y_t) \sim \text{Unif}(\mathcal{M}_t) \\
 S_{t+1} &= \Pi_{[0,1]} \left[S_t + \beta_t \cdot (\text{SCXUpdate}(S_t, \mathcal{M}_{t+1}, f_{\theta_{t+1}}) - S_t) \right]
 \end{aligned}$$

where $\gamma \in (0, 1)$ is the acceptance threshold, α_t is the student learning rate, β_t is the gatekeeper update rate, and $\Pi_{[0,1]}$ projects scores to $[0, 1]$.

Reference-set evaluation (C10). The gatekeeper update $\text{SCXUpdate}(S_t, \mathcal{M}_{t+1}, f_{\theta_{t+1}})$ is evaluated on a fixed reference set $\mathcal{M}_0$ (see condition C10), not on the growing memory bank. This anchors the evaluation to a stationary distribution and is required for Lyapunov descent (Lemma SE-1.1). Without reference-set replay, the gatekeeper can trivially reduce its loss by accepting only easy samples (Theorem 12.2).

Assume the following conditions hold:

(C1) Finite structure space. The input space $\mathcal{X}$ is finite or has finite covering dimension: $\dim(\mathcal{X}) < \infty$.

> **FIXED (2026-06-28, DEFECT-15):** The original “finite $\mathcal{X}$ ” assumption is violated in practice (continuous $\mathbb{R}^{d_\phi}$ feature spaces). This section replaces it with a **covering number argument** that achieves the same memory bank stabilization result (Lemma SE-1.2) under the much weaker condition of finite covering dimension. See Section 3.1 for the full construction.

(C2) Lipschitz student. The NEP student f_θ is Lipschitz continuous in θ with constant L_f :

$$\|f_{\theta_1}(x) - f_{\theta_2}(x)\| \leq L_f \|\theta_1 - \theta_2\|, \quad \forall x \in \mathcal{X}, \forall \theta_1, \theta_2$$

(C3) Lipschitz gatekeeper. The SCX update function is Lipschitz continuous with constant L_S :

$$\|\text{SCXUpdate}(S_1, \mathcal{M}, f) - \text{SCXUpdate}(S_2, \mathcal{M}, f)\|_\infty \leq L_S \|S_1 - S_2\|_\infty$$

(C4) Learning rates. The student learning rate α_t satisfies the Robbins-Monro conditions:

$$\sum_{t=1}^{\infty} \alpha_t = \infty, \quad \sum_{t=1}^{\infty} \alpha_t^2 < \infty$$

The gatekeeper update rate β_t satisfies:

$$\beta_t \rightarrow 0, \quad \sum_{t=1}^{\infty} \beta_t^2 < \infty$$

FIXED (2026-06-28, B8): The original C4 listed $\sum \beta_t = \infty$ (Robbins-Monro exploration) for the gatekeeper. However, for the cumulative distribution shift to be finite (needed for Lyapunov descent), the stronger condition $\sum_{t=1}^{\infty} \beta_t < \infty$ is required. This is now stated as an **explicit separate sub-condition (C4b)** — it does NOT follow from $\beta_t = o(\alpha_t)$ (C6) combined with $\sum \alpha_t^2 < \infty$. See the counterexample in the C6 discussion below.

Resolution of the tension: The gatekeeper exploration (originally $\sum \beta_t = \infty$) is instead provided by the random exploration

fraction (C9) and the annealing acceptance threshold (C8). The gatekeeper update rate itself must satisfy $\sum \beta_t < \infty$ to ensure finite total distribution shift. The canonical schedule is updated to $\alpha_t = t^{-a}$, $\beta_t = t^{-b}$ with $\frac{1}{2} < a < 1 < b$, e.g., $a = 0.6$, $b = 1.2$.

(C5) Conditional i.i.d. sampling. Conditioned on S_t , the samples drawn at time t are i.i.d. from the acceptance-biased distribution:

$$(x_t, y_t) \mid S_t \sim P_{S_t}(x, y) \propto S_t(x, y) \cdot P_0(x, y)$$

(C6) Two-timescale separation (corrected — replaces original “sufficient annealing”). The gatekeeper update rate β_t must be asymptotically slower than the student learning rate α_t :

$$\boxed{\beta_t = o(\alpha_t)}, \quad \text{i.e.,} \quad \lim_{t \rightarrow \infty} \frac{\beta_t}{\alpha_t} = 0.$$

Rationale (DEFECT-14 fix). The original condition C6 ($\max(\alpha_t, \beta_t) \rightarrow 0$) is insufficient because it does not resolve the tension between gatekeeper exploration ($\sum \beta_t = \infty$) and student stabilization (requires approximately stationary distribution). The two-timescale condition $\beta_t = o(\alpha_t)$ resolves this: between any two gatekeeper updates, the student takes asymptotically many gradient steps, ensuring local convergence to the current distribution’s minimum before the distribution shifts.

Jointly satisfiable scheduling scheme (corrected B8). The conditions $\sum \alpha_t = \infty$, $\sum \alpha_t^2 < \infty$, $\sum \beta_t < \infty$, $\sum \beta_t^2 < \infty$, and $\beta_t = o(\alpha_t)$ are jointly satisfiable. A canonical example is:

$$\alpha_t = t^{-a}, \quad \beta_t = t^{-b}, \quad \text{with} \quad \frac{1}{2} < a < 1 < b.$$

For $a = 0.6$, $b = 1.2$: - $\sum \alpha_t = \infty$ (harmonic series with exponent < 1) - $\sum \alpha_t^2 = \sum t^{-1.2} < \infty$ (Robbins-Monro for student) - $\sum \beta_t = \sum t^{-1.2} < \infty$ (finite total distribution shift — explicit condition) - $\sum \beta_t^2 = \sum t^{-2.4} < \infty$ (variance control for gatekeeper) - $\beta_t/\alpha_t = t^{-0.6} \rightarrow 0$ (two-timescale separation)

Consequence of violation. Without two-timescale separation (β_t not $o(\alpha_t)$), the cumulative distribution shift may diverge:

$$\sum_{t=1}^{\infty} TV(P_{t+1}, P_t) \leq \sum_{t=1}^{\infty} \frac{2\beta_t B_S}{\mathbb{E}_{P_0}[S_t] - \beta_t B_S}.$$

FIXED (2026-06-28, B8): $\sum \beta_t < \infty$ is a separate condition, **NOT implied by** $\beta_t = o(\alpha_t)$. The original text claimed that $\beta_t = o(\alpha_t)$ combined with $\sum \alpha_t^2 < \infty$ implies $\sum \beta_t < \infty$. This is **mathematically false**, as shown by the following counterexample:

$$\alpha_t = t^{-0.6}, \quad \beta_t = \frac{1}{t^{0.6} \log t}.$$

Then: - $\beta_t/\alpha_t = 1/\log t \rightarrow 0$, so $\beta_t = o(\alpha_t)$ - $\sum \alpha_t^2 = \sum t^{-1.2} < \infty$ - But $\sum \beta_t = \sum \frac{1}{t^{0.6} \log t} = \infty$ (diverges by the integral test)

The flaw: $\beta_t = o(\alpha_t)$ only guarantees $\beta_t < \alpha_t$ for large t , but $\sum \alpha_t^2 < \infty$ controls the **squares** of α_t , not α_t itself. Since $\sum \alpha_t = \infty$ (Robbins-Monro), α_t decays as t^{-a} with $a \in (1/2, 1]$, and $\beta_t = o(\alpha_t)$ only forces $b > a$, not $b > 1$ (which is what $\sum \beta_t < \infty$ requires).

Resolution. The condition $\sum_{t=1}^{\infty} \beta_t < \infty$ is now stated as an **explicit separate requirement** in C4 (below), independent of the two-timescale condition C6. It must be verified independently or enforced through the

specific schedule $\beta_t = t^{-b}$ with $b > 1$ (which is stronger than the Robbins-Monro $\sum \beta_t = \infty$ originally listed in C4 — see the corrected C4 below).

Under the corrected C4 (which now requires $\sum \beta_t < \infty$ for the gatekeeper, distinct from the student's $\sum \alpha_t = \infty$), the total distribution shift is finite and the student sees an asymptotically stationary distribution.

(C7) Gatekeeper bounded gradient. The SCX update increment is bounded:

$$\|\text{SCXUpdate}(S, \mathcal{M}, f) - S\|_\infty \leq B_S < \infty$$

(C8) Annealing acceptance threshold (sufficient, not necessary).

The acceptance threshold γ_t starts low (permissive, $\gamma_0 < 0.5$) and increases to 0.5 as $t \rightarrow \infty$:

$$\gamma_t = \gamma_0 + (0.5 - \gamma_0)(1 - e^{-t/\tau_\gamma}).$$

This prevents early over-confidence while allowing increasing selectivity as the gatekeeper improves.

(C9) Random exploration fraction (sufficient, not necessary). A

fixed fraction $\varepsilon_{\text{explore}} > 0$ of accepted samples are chosen uniformly at random regardless of S_t score, ensuring all regions remain represented in $\mathcal{M}_t$.

B7 FIX — $\varepsilon > 0$ lower bound for importance weights:

The importance weight bound $\|w\|_\infty \leq W < \infty$ in Theorem 12.5 requires $P_{S_t}(x) > 0$ for all x in the support of P_0 , i.e., $\inf_{x \in \text{supp}(P_0)} P_{S_t}(x) > 0$. When $\varepsilon_{\text{explore}} = 0$ (no exploration), the gatekeeper can zero out $S_t(x)$ for regions it deems unreli-

able, making $P_{S_t}(x) = 0$ and $w(x) = P_0(x)/P_{S_t}(x) = \infty$ — the importance weights become unbounded.

Resolution: Condition C9 requires $\varepsilon_{\text{explore}} > 0$ **strictly**. With $\varepsilon_{\text{explore}} > 0$, every x with $P_0(x) > 0$ has $P_{S_t}(x) \geq \varepsilon_{\text{explore}} \cdot P_0(x) > 0$ (the uniform exploration component ensures non-zero probability). Consequently:

$$\|w\|_\infty = \sup_x \frac{P_0(x)}{P_{S_t}(x)} \leq \frac{1}{\varepsilon_{\text{explore}}} = W < \infty.$$

The bound W is finite and explicit: $W = 1/\varepsilon_{\text{explore}}$. As $\varepsilon_{\text{explore}} \rightarrow 0$, $W \rightarrow \infty$, and the variance of the importance-weighted gradient estimator diverges — confirming that $\varepsilon_{\text{explore}} > 0$ is a **necessary condition** for the Lyapunov descent claim in Theorem 12.5 to be non-vacuous.

Note on C8-C9: These are sufficient conditions to mitigate the selection bias cycle (§5.1). They are not mathematically necessary for the theorem statement, but without them (or equivalent mechanisms), the Lyapunov descent claim in Lemma SE-1.1 is not guaranteed — the decrease in V_t on P_{S_t} may reflect distribution shift rather than genuine improvement.

(C10) Reference-set replay (REQUIRED for Lyapunov descent). A fixed reference set $\mathcal{M}_0$ and validation set $\mathcal{V}_0$ are held out before self-evolution begins. The Lyapunov function Ψ is evaluated on these fixed sets, NOT on the growing memory bank $\mathcal{M}_t$:

$$\Psi(S_t, \theta_t) = \frac{1}{|\mathcal{M}_0|} \sum_{x \in \mathcal{M}_0} (S_t(x, y(x)) - \hat{C}(x))^2 + \frac{\lambda}{|\mathcal{V}_0|} \sum_{(x, y) \in \mathcal{V}_0} \ell(f_{\theta_t}(x), y).$$

Critical dependency: Condition C10 is the bridge between Lemma SE-1.1 and Theorems 12.2/12.5. Theorem 12.2 proves that Lyapunov descent is **formally impossible** without reference-set replay (the gatekeeper can trivially reduce loss by accepting only easy samples). Theorem 12.5 proves that **with** reference-set replay and importance sampling, Lyapunov descent is achievable. Lemma SE-1.1 is therefore **conditional on C10** — its “CONDITIONAL” status in the table below assumes C10 holds. **Without replay, Lemma SE-1.1 is not guaranteed (see Thm 12.2).**

Without C10, Lemma SE-1.1’s supermartingale argument is valid in form, but the Lyapunov function $V_t = \mathbb{E}_{P_{S_t}}[\ell(f_{\theta_t}(x), y)]$ measures performance on the acceptance-biased distribution, which can decrease even as true performance degrades (the selection bias cycle, §5.1). C10 breaks this cycle by anchoring evaluation to a fixed reference.

3. Conditions for Theorem SE-1

3.1 Condition C1: Finite Structure Space — Covering Number Replacement (DEFECT-15 Fix)

Original (defective): C1 assumed $\mathcal{X}$ is finite. This is violated in practice for continuous feature spaces $\phi : \mathcal{X} \rightarrow \mathbb{R}^{d_\phi}$, where the input space is a continuum and the memory bank can grow without bound in terms of distinguishable configurations.

FIXED (2026-06-28): Covering Number Argument. We replace the finite- $\mathcal{X}$ assumption with a covering-number construction that achieves the same stabilization result under the much weaker condition of finite covering dimension.

Construction (-net partitioning):

1. **Finite covering dimension.** Assume the feature space $\Phi = \phi(\mathcal{X}) \subseteq \mathbb{R}^{d_\phi}$ has finite Euclidean dimension $d_\phi < \infty$ (always true for any practical feature map).
2. **Lipschitz gatekeeper (C3).** The SCX score function $S_t(x, y)$ is L_S -Lipschitz in x by condition C3. This means: within any ε/L_S -ball in feature space, S_t varies by at most ε .
3. **-net construction.** Partition Φ into N_ε hypercubes of side ε/L_S :

$$N_\varepsilon = \left\lceil \frac{\text{diam}(\Phi) \cdot L_S}{\varepsilon} \right\rceil^{d_\phi} = O\left(\left(\frac{L_S}{\varepsilon}\right)^{d_\phi}\right)$$

4. **Finite distinguishable types.** Within each cube, all points are ε -equivalent in gatekeeper score. The memory bank state is entirely determined by **which cubes have been visited and accepted**. Since there are N_ε cubes and each is either visited-or-not, the number of possible memory bank configurations is at most:

$$|\mathcal{M}\text{-configs}| \leq 2^{N_\varepsilon} = \exp\left(O\left(\left(\frac{L_S}{\varepsilon}\right)^{d_\phi}\right)\right)$$

which is **finite** for any fixed $\varepsilon > 0$.

5. **Stabilization up to -precision.** Lemma SE-1.2's finite-cover argument now applies: if two memory bank configurations are ε -close (same cubes visited), the gatekeeper scores S_t differ by at most $O(\varepsilon)$. Since

machine precision is finite ($\varepsilon_{\text{mach}} \approx 10^{-16}$ for float64), taking $\varepsilon = \varepsilon_{\text{mach}}$ yields a genuinely finite configuration space for all practical purposes.

6. Infinite-space extension (sketch). For the limit $\varepsilon \rightarrow 0$, the configuration space grows as $2^{O((1/\varepsilon)^{d_\phi})}$, which is doubly exponential in $1/\varepsilon$. However, the stabilization result only requires that **for any given precision $\varepsilon > 0$, the system enters and stays within an ε -neighborhood of a fixed point after finite time.** This follows because:

- The Lyapunov descent (conjectured) drives the system toward the fixed point
- Once within distance ε , the ε -net only needs to resolve $O((1/\varepsilon)^{d_\phi})$ cubes
- Finite-time convergence to ε -precision holds for each fixed ε

Practical interpretation. This covering-number construction replaces the unrealistic “finite $\mathcal{X}$ ” assumption with the genuinely weak “finite covering dimension” assumption, which holds for **any** feature space embedded in $\mathbb{R}^{d_\phi}$ — including continuous pixel spaces, molecular descriptor spaces, and language model embedding spaces. The same stabilization result (Lemma SE-1.2) holds, now justified by the Lipschitz property (C3) and the finite-dimensional feature embedding rather than by an artificial finiteness constraint.

Remaining gap. The infinite-space limit $\varepsilon \rightarrow 0$ with full functional convergence (rather than ε -precision approximate convergence) requires additional compactness conditions (e.g., Arzelà-Ascoli for the space of score functions) that are beyond the current scope. This is noted as an open problem in the verification report.

3.2 Conditions C2-C3: Lipschitz Continuity

These conditions ensure that small changes in parameters or gatekeeper scores produce proportionally small changes in the system state.

3.2.1 Lemma SE-1.0: Indicator Discontinuity Is Measure-Zero (B9 Fix) Lemma SE-1.0 (Lipschitz-Almost-Everywhere Consensus).

Under condition C5 (conditional i.i.d. sampling), the consensus score $C(x) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\ell(f_m(x), y) > \tau\}$ is Lipschitz-almost-everywhere with respect to the acceptance-biased data distribution P_{S_t} . Specifically, the set of points where C is discontinuous has P_{S_t} -measure zero:

$$\mathbb{P}_{(x,y) \sim P_{S_t}}(\exists m : \ell(f_m(x), y) = \tau) = 0.$$

Why this resolves the B9 gap. The ARXIV verification identified a fundamental gap: condition C3 requires the gatekeeper and its update to be Lipschitz continuous, but the SCX consensus computation $C(x)$ uses indicator functions $\mathbf{1}\{\ell > \tau\}$ that are discontinuous at the threshold τ . However:

1. **Measure-zero under C5.** Since $P_{S_t} \propto S_t \cdot P_0$ is absolutely continuous with respect to P_0 , and under the mild assumption that $\ell(f_m(x), y)$ has a continuous distribution (true when the student f_θ is a smooth function — e.g., any neural network with continuous activations — and the data distribution has a density), the event $\ell(f_m(x), y) = \tau$ has probability zero. The discontinuities occur on a set of Lebesgue measure zero.
2. **Lipschitz-almost-everywhere is sufficient.** The Lyapunov analysis in Lemma SE-1.1 uses expectations $\mathbb{E}_{P_{S_t}}[\cdot]$, and expectations are

invariant to modifications on measure-zero sets. The standard theory of stochastic approximation (Borkar, 2008; Kushner & Yin, 2003) requires Lipschitz continuity of the mean field, not pointwise continuity. The mean field $\mathbb{E}_{P_{S_t}}[C(x)]$ is Lipschitz continuous even when $C(x)$ has jump discontinuities on a null set, as long as the distribution has a density.

3. **Sigmoid smoothing as optional refinement.** For implementations concerned with gradient-based optimization through $C(x)$, the indicator can be replaced by a smooth sigmoid $\sigma(\beta(\ell - \tau))$ with $\beta \gg 1$. The approximation error satisfies $\|\mathbf{1}\{\ell > \tau\} - \sigma(\beta(\ell - \tau))\|_1 = O(1/\beta)$. Taking $\beta \rightarrow \infty$ recovers the hard threshold, and the proof chain remains valid because (a) the expectation converges uniformly in β , and (b) the Lipschitz constant of the sigmoid-smoothed consensus is $O(\beta)$, which can be absorbed into the step-size conditions via the two-timescale separation (C6).

Formal justification (measure-zero argument):

Proof of Lemma SE-1.0. For each expert m , the student loss $\ell(f_m(x), y)$ is a random variable induced by the sampling process. Under C5, (x, y) is drawn from P_{S_t} , which has a density with respect to the base measure. The function $g_m(x, y) = \ell(f_m(x), y)$ is continuous in (x, y) (composition of the continuous student f_{θ_t} and the continuous loss ℓ). For a continuous random variable, the probability of hitting any specific value τ is zero: $\mathbb{P}(g_m(X, Y) = \tau) = 0$. By the union bound over the M experts:

$$\mathbb{P}(\exists m : \ell(f_m(X), Y) = \tau) \leq \sum_{m=1}^M \mathbb{P}(\ell(f_m(X), Y) = \tau) = 0.$$

Therefore, $C(x)$ is almost surely continuous — its discontinuity set (the union

of the M level sets $\{\ell(f_m(x), y) = \tau\}$ has probability zero. On the complement of this null set, $C(x)$ is locally constant (piecewise constant with jumps only at the threshold crossings, which occur with probability zero). Hence C is Lipschitz-almost-everywhere (trivially, with Lipschitz constant 0 on each continuity region). $\square$

Implications for the proof chain: - The Lyapunov descent (Lemma SE-1.1) uses $\mathbb{E}[\cdot]$, which is unchanged by null-set modifications. - The SCX update $\text{SCXUpdate}(S_t, \mathcal{M}_t, f_{\theta_t})$ inherits Lipschitz-almost-everywhere continuity because it aggregates $C(x)$ over finite memory banks. - The stochastic approximation theory applies to the mean field, which is Lipschitz continuous. - Therefore: **the indicator discontinuity does NOT break the proof chain.** The gap identified in the ARXIV verification is closed by this lemma.

Proposition SE-1.0 (Lipschitz Composition). Under C2 and C3, the student loss function is Lipschitz in θ with composite constant $L_\ell \cdot L_f$, where L_ℓ is the Lipschitz constant of the loss ℓ :

$$|\ell(f_{\theta_1}(x), y) - \ell(f_{\theta_2}(x), y)| \leq L_\ell L_f \|\theta_1 - \theta_2\|$$

and the gradient is bounded by $G = L_\ell L_f$.

3.3 Conditions C4-C6: Learning Rates and Sampling

Condition C4 ensures enough exploration ($\sum \alpha_t = \infty$) while controlling noise ($\sum \alpha_t^2 < \infty$). Condition C5 ensures that the memory bank $\mathcal{M}_t$ is a representative sample of the acceptance-biased distribution. Condition C6 ensures that the system eventually stabilizes rather than oscillating indefinitely due to non-vanishing step sizes.

3.4 Condition C7: Bounded Gatekeeper Update

This ensures that the gatekeeper scores remain in $[0, 1]$ and that the update increments are integrable. The projection $\Pi_{[0,1]}$ enforces this explicitly.

4. Conclusion of Theorem SE-1

Under conditions (C1)-(C7), the sequence (S_t, θ_t) converges to a **fixed point** (S^*, θ^*) almost surely, where:

(a) Self-consistent gatekeeper. S^* is self-consistent in the Bayesian sense:

$$S^*(x, y) = P(\text{clean} \mid x, y, \mathcal{M}_\infty), \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y}$$

where $\mathcal{M}_\infty = \lim_{t \rightarrow \infty} \mathcal{M}_t$ is the limiting memory bank (possibly infinite). Equivalently, S^* satisfies the fixed-point equation:

$$S^* = \text{SCXUpdate}(S^*, \mathcal{M}_\infty, f_{\theta^*})$$

(b) Optimal NEP student (local). θ^* is a local minimum of the expected NEP loss under the limiting data distribution induced by S^* :

$$\theta^* = \arg \min_{\theta \in \Theta} \mathbb{E}_{(x,y) \sim P_{S^*}} [\ell(f_\theta(x), y)] \quad (\text{local})$$

where $P_{S^*}(x, y) \propto S^*(x, y) \cdot P_0(x, y)$ is the acceptance-biased distribution at the fixed point.

(c) **Self-consistency equation.** The fixed point satisfies the coupled system:

$$\begin{aligned} S^* &= \mathcal{T}_S(S^*, \theta^*) \\ \theta^* &= \mathcal{T}_\theta(S^*, \theta^*) \end{aligned}$$

where $\mathcal{T}_S$ and $\mathcal{T}_\theta$ are the gatekeeper and student update operators respectively, evaluated at the fixed point.

(d) **Lyapunov convergence.** The total loss function $V(S, \theta) = \mathbb{E}_{(x,y) \sim P_S}[\ell(f_\theta(x), y)]$ serves as a Lyapunov function that is strictly decreasing along the trajectory until the fixed point is reached.

5. Lemma SE-1.1: Lyapunov Function Non-Increase

Lemma SE-1.1 (Lyapunov Function). Define the total system loss:

$$V_t = V(S_t, \theta_t) = \mathbb{E}_{(x,y) \sim P_{S_t}}[\ell(f_{\theta_t}(x), y)] + \lambda \cdot \|S_t - S_{\text{prior}}^*\|_{\mathcal{L}^2}^2$$

where $\lambda > 0$ regularizes the gatekeeper toward a prior S_{prior}^* (e.g., the initial SCX score), and P_S is the acceptance-biased distribution under gatekeeper S .

Under conditions (C2)-(C7) and **C10** (reference-set replay — REQUIRED for Lyapunov descent; without C10 Lemma SE-1.1 is NOT guaranteed, see Theorem 12.2):

$$\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \leq V_t - \eta_t$$

where $\eta_t \geq 0$ is a non-negative random variable satisfying:

$$\eta_t = \alpha_t \cdot \|\nabla_{\theta} L_t(\theta_t)\|^2 + \beta_t \cdot \|\Delta S_t\|_{\mathcal{L}^2}^2 + o(\alpha_t + \beta_t)$$

with $\Delta S_t = \text{SCXUpdate}(S_t, \mathcal{M}_t, f_{\theta_t}) - S_t$.

Proof sketch.

1. **Decompose the change.** The total change in V_t has three components:
 - Change due to θ update: $\Delta_{\theta} V_t = V(S_t, \theta_{t+1}) - V(S_t, \theta_t)$
 - Change due to S update: $\Delta_S V_t = V(S_{t+1}, \theta_{t+1}) - V(S_t, \theta_{t+1})$
 - Cross terms: $o(\alpha_t + \beta_t)$
2. **Student contribution.** From the Lipschitz gradient assumption (SEA7):

$$\mathbb{E}[\Delta_{\theta} V_t \mid \mathcal{F}_t] \leq -\alpha_t \|\nabla_{\theta} L_t(\theta_t)\|^2 + \frac{L_g \alpha_t^2}{2} G^2$$

3. **Gatekeeper contribution.** From the Lipschitz property of SCXUpdate:

$$\mathbb{E}[\Delta_S V_t \mid \mathcal{F}_t] \leq -\beta_t (\|\Delta S_t\|_{\mathcal{L}^2}^2 - 2\lambda \langle \Delta S_t, S_t - S_{\text{prior}}^* \rangle) + O(\beta_t^2)$$

By the Cauchy-Schwarz inequality and the bound $\|\Delta S_t\|_{\infty} \leq B_S$, the cross term is $O(\beta_t)$.

4. **Combining terms.** Under sufficient annealing ($\beta_t \rightarrow 0, \alpha_t \rightarrow 0$), the negative quadratic terms dominate the $O(\alpha_t^2 + \beta_t^2)$ corrections, giving the supermartingale inequality.

Corollary SE-1.1 (Convergence of V_t). The Lyapunov sequence $\{V_t\}$ converges almost surely to a finite limit V_∞ :

$$V_t \xrightarrow{a.s.} V_\infty \geq 0$$

and moreover:

$$\sum_{t=1}^{\infty} (\alpha_t \|\nabla_{\theta} L_t(\theta_t)\|^2 + \beta_t \|\Delta S_t\|_{\mathcal{L}^2}^2) < \infty \quad \text{a.s.}$$

5.1 Selection Bias Cycle Analysis (DEFECT-13 Fix)

A critical concern for the Lyapunov descent proof is the **selection bias cycle** inherent in the self-evolution loop:

$$\text{Gatekeeper } S_t \xrightarrow{\text{selects}} \text{Memory } \mathcal{M}_{t+1} \xrightarrow{\text{trains}} \text{NEP } f_{\theta_{t+1}} \xrightarrow{\text{feedback}} \text{Gatekeeper } S_{t+1}$$

The problem. At each iteration, the gatekeeper S_t selects which samples to admit to $\mathcal{M}_{t+1}$ based on its current scoring function. The NEP student is then trained on this acceptance-biased dataset. The NEP's outputs feed back to update the gatekeeper. This creates a self-reinforcing cycle:

1. If S_t is overly confident about certain regions of $\mathcal{X}$, it will admit samples predominantly from those regions, excluding counter-examples.

2. The NEP student, trained on this biased subset, will perform well on the admitted regions but may degrade on excluded regions.
3. The updated gatekeeper S_{t+1} receives feedback from a student that is biased toward the accepted distribution, reinforcing the original bias.

Formalization. Let P_0 be the original data distribution and P_{S_t} be the acceptance-biased distribution:

$$P_{S_t}(x, y) \propto \mathbf{1}\{S_t(x, y) \geq \gamma\} \cdot P_0(x, y).$$

The Lyapunov function $V_t = \mathbb{E}_{(x,y) \sim P_{S_t}} [\ell(f_{\theta_t}(x), y)]$ measures performance on the **acceptance-biased** distribution, not on the original distribution P_0 . A decrease in V_t does **not** guarantee improvement on P_0 — it may simply reflect the gatekeeper becoming better at selecting easy samples.

Quantifying the distribution shift. The total variation between the original and acceptance-biased distributions at time t is:

$$TV(P_0, P_{S_t}) = \frac{1}{2} \sum_{x,y} |P_0(x, y) - P_{S_t}(x, y)| \leq \mathbb{P}_{P_0}(S_t(x, y) < \gamma).$$

As the gatekeeper becomes more selective (higher γ_t or stricter S_t), the distribution shift can grow, making the Lyapunov decrease on P_{S_t} increasingly disconnected from true performance on P_0 .

Mitigation strategies (proposed for Conjecture SE-1):

1. **Reference-set evaluation (DEFECT-03/04 fix).** Evaluate both gatekeeper and student on a fixed reference set M_0 (held out before

self-evolution begins), not on the growing memory bank:

$$\Phi_{\text{ref}}(S_t, \theta_t) = \frac{1}{|M_0|} \sum_{x \in M_0} (S_t(x, y(x)) - \hat{C}(x))^2 + \frac{\lambda}{|V_0|} \sum_{(x, y) \in V_0} \ell(f_{\theta_t}(x), y).$$

2. **Random exploration (C9).** A fixed fraction $\varepsilon > 0$ of accepted samples are chosen uniformly at random regardless of S_t score, ensuring all regions remain represented in $\mathcal{M}_t$.
3. **Annealing acceptance threshold (C8).** The threshold γ_t starts low (permissive, $\gamma_0 < 0.5$) and increases to 0.5 as $t \rightarrow \infty$:

$$\gamma_t = \gamma_0 + (0.5 - \gamma_0)(1 - e^{-t/\tau_\gamma}).$$

This prevents early over-confidence while allowing increasing selectivity as the gatekeeper improves.

4. **Two-timescale separation (DEFECT-14 fix).** Gatekeeper updates are asymptotically slower than student updates ($\beta_t = o(\alpha_t)$), ensuring the student converges to a local minimum on the current distribution before the distribution shifts significantly.

Degradation under bias accumulation. If the selection bias is not controlled, the effective information available to the system degrades. Let $\mathcal{R}_{\text{excluded}}(t) = \{x : S_t(x, y) < \gamma_t\}$ be the set of excluded samples. The student's generalization error on the full distribution P_0 satisfies:

$$\mathbb{E}_{P_0}[\ell(f_{\theta_t}(x), y)] = \mathbb{E}_{P_{S_t}}[\ell(f_{\theta_t}(x), y)] \cdot \mathbb{P}(S_t \geq \gamma_t) + \mathbb{E}_{\mathcal{R}_{\text{excluded}}(t)}[\ell(f_{\theta_t}(x), y)] \cdot \mathbb{P}(S_t < \gamma_t).$$

The second term is unobserved if no exploration is performed, and can grow without detection.

Status. The selection bias cycle is the primary obstacle to proving Lyapunov descent. The mitigation strategies (C8, C9) are **conjectured** to be sufficient, but a rigorous proof that they break the self-reinforcing cycle remains open. Without these mitigations, the Lyapunov decrease on P_{S_t} is **not** a valid indicator of system improvement — it may reflect distribution shift, not genuine model improvement.

6. Lemma SE-1.2: Finite Cover of Memory Bank Types

Lemma SE-1.2 (Finite Memory Bank Configurations). Under condition C1 (finite $\mathcal{X}$) and the acceptance criterion $S_t(x, y) \geq \gamma$:

- (a) The number of possible memory bank configurations is finite.
- (b) The memory bank $\mathcal{M}_t$ evolves monotonically ($\mathcal{M}_t \subseteq \mathcal{M}_{t+1}$) and stabilizes after finitely many additions:

$$\mathcal{M}_t = \mathcal{M}_{t+1} = \dots = \mathcal{M}_\infty \quad \text{for all sufficiently large } t$$

- (c) Consequently, a.s. there exists a finite time T such that for all $t \geq T$:

$$\Delta\mathcal{M}_t = \mathcal{M}_{t+1} \setminus \mathcal{M}_t = \emptyset$$

Proof sketch.

1. **Finiteness.** Since $\mathcal{X}$ is finite, $\mathcal{X} \times \mathcal{Y}$ is finite. The memory bank $\mathcal{M}_t$ is a subset of $\mathcal{X} \times \mathcal{Y}$. There are at most $2^{|\mathcal{X} \times \mathcal{Y}|}$ possible memory banks, which is finite.

2. **Monotonicity.** By definition, $\mathcal{M}_{t+1} = \mathcal{M}_t \cup \text{accepted}_t$, so $\mathcal{M}_t$ is monotonically increasing ($\mathcal{M}_t \subseteq \mathcal{M}_{t+1}$).
3. **Stabilization.** A monotonically increasing sequence in a finite set must stabilize after finitely many steps. Therefore there exists T such that $\mathcal{M}_t = \mathcal{M}_{t+1}$ for all $t \geq T$.

Corollary SE-1.2 (Infinite $\mathcal{X}$ Extension). If $\mathcal{X}$ is infinite but has finite covering dimension, the number of “types” (equivalence classes of configurations under the SCX score topology) is finite. The stabilization result holds up to ε -equivalence: for any $\varepsilon > 0$, there exists T_ε such that:

$$\sup_{x,y} |S_t(x,y) - S_{t'}(x,y)| < \varepsilon \quad \text{for all } t, t' \geq T_\varepsilon$$

This extension requires the additional assumption that the SCX score function is uniformly continuous in the data distribution.

7. Lemma SE-1.3: Parameter Displacement Vanishes

Lemma SE-1.3 (Vanishing Parameter Displacement). Under conditions (C2)-(C7):

$$\|\theta_{t+1} - \theta_t\| \xrightarrow{a.s.} 0 \quad \text{and} \quad \|S_{t+1} - S_t\|_\infty \xrightarrow{a.s.} 0$$

as $t \rightarrow \infty$.

Proof sketch.

1. **Student displacement.** From the update rule and bounded gradient (SE-A8):

$$\|\theta_{t+1} - \theta_t\| = \alpha_t \|g_t(\theta_t)\| \leq \alpha_t G$$

Since $\alpha_t \rightarrow 0$ (by C6), $\|\theta_{t+1} - \theta_t\| \rightarrow 0$.

2. **Gatekeeper displacement.** From the update rule and bounded update (C7):

$$\|S_{t+1} - S_t\|_\infty = \beta_t \|\text{SCXUpdate}(S_t, \mathcal{M}_t, f_{\theta_t}) - S_t\|_\infty \leq \beta_t B_S$$

Since $\beta_t \rightarrow 0$ (by C6), $\|S_{t+1} - S_t\|_\infty \rightarrow 0$.

Corollary SE-1.3 (Joint Convergence). The sequence (S_t, θ_t) is Cauchy in the product space $\mathcal{L}^\infty(\mathcal{X} \times \mathcal{Y}) \times \mathbb{R}^{d_\theta}$ with probability 1:

$$\lim_{t \rightarrow \infty} \sup_{t' \geq t} (\|S_{t'} - S_t\|_\infty + \|\theta_{t'} - \theta_t\|) = 0 \quad \text{a.s.}$$

8. Lemma SE-1.4: Limit Points Are Fixed Points

Lemma SE-1.4 (Fixed Point Characterization). Let $(S_\infty, \theta_\infty)$ be any limit point of the sequence (S_t, θ_t) (along a subsequence $t_k \rightarrow \infty$). Then:

- (a) **Gatekeeper fixed point.** S_∞ satisfies:

$$S_\infty = \text{SCXUpdate}(S_\infty, \mathcal{M}_\infty, f_{\theta_\infty})$$

(b) **Student fixed point.** θ_∞ satisfies:

$$\nabla_{\theta} \mathbb{E}_{(x,y) \sim P_{S_\infty}} [\ell(f_{\theta_\infty}(x), y)] = 0$$

(c) **Cross-consistency.** The pair $(S_\infty, \theta_\infty)$ is a joint fixed point of the coupled system:

$$\begin{aligned} S_\infty &= \mathcal{T}_S(S_\infty, \theta_\infty) \\ \theta_\infty &= \mathcal{T}_\theta(S_\infty, \theta_\infty) \end{aligned}$$

Proof sketch.

1. **Gatekeeper fixed point.** From Lemma SE-1.1 and Corollary SE-1.1:

$$\sum_{t=1}^{\infty} \beta_t \|\Delta S_t\|_{\mathcal{L}^2}^2 < \infty \quad \text{a.s.}$$

Since $\sum \beta_t = \infty$ (C4), it must be that $\|\Delta S_t\|_{\mathcal{L}^2} \rightarrow 0$ along a subsequence with $\beta_t > 0$. By the Lipschitz continuity of SCXUpdate (C3):

$$\|S_\infty - \text{SCXUpdate}(S_\infty, \mathcal{M}_\infty, f_{\theta_\infty})\|_\infty = 0$$

2. **Student fixed point.** From Lemma SE-1.1 and Corollary SE-1.1:

$$\sum_{t=1}^{\infty} \alpha_t \|\nabla_{\theta} L_t(\theta_t)\|^2 < \infty \quad \text{a.s.}$$

Since $\sum \alpha_t = \infty$ (C4), $\|\nabla_{\theta} L_t(\theta_t)\| \rightarrow 0$ along a subsequence. By Lemma SE-1.3 and the continuity of ∇L_∞ , convergence of θ_t implies $\nabla_{\theta} L_\infty(\theta_\infty) = 0$.

3. **Cross-consistency.** From Lemma SE-1.2, $\mathcal{M}_t$ stabilizes at $\mathcal{M}_\infty$. At the limit point, S_∞ induces P_{S_∞} , θ_∞ minimizes the loss under P_{S_∞} , and S_∞ is the Bayes estimate under $\mathcal{M}_\infty$. These three conditions are mutually consistent.
-

9. Proof Sketch of Theorem SE-1

Here we assemble the lemmas into the complete proof of Theorem SE-1.

Step 1: Lyapunov descent (Lemma SE-1.1). Show that $V_t = V(S_t, \theta_t)$ is a supermartingale with non-negative drift. From the update rules, the one-step expected change in V_t is bounded above by $-\eta_t$ where $\eta_t \geq 0$ measures progress. By the supermartingale convergence theorem, $V_t \rightarrow V_\infty$ a.s. and $\sum \eta_t < \infty$ a.s.

Step 2: Memory bank stabilization (Lemma SE-1.2). Because $\mathcal{X}$ is finite (C1), the memory bank $\mathcal{M}_t$ can take only finitely many values. Since it is monotonically increasing ($\mathcal{M}_t \subseteq \mathcal{M}_{t+1}$), it must stabilize after finitely many steps. Thus there exists $T < \infty$ such that $\mathcal{M}_t = \mathcal{M}_\infty$ for all $t \geq T$.

Step 3: Displacement vanishes (Lemma SE-1.3). From the annealing condition (C6), $\alpha_t \rightarrow 0$ and $\beta_t \rightarrow 0$, so successive iterates are arbitrarily close:

$$\|\theta_{t+1} - \theta_t\| \leq \alpha_t G \rightarrow 0, \quad \|S_{t+1} - S_t\|_\infty \leq \beta_t B_S \rightarrow 0$$

Step 4: Convergence of parameter sequence. The Lyapunov descent (Step 1) combined with vanishing displacement (Step 3) implies that the

sequence (S_t, θ_t) is Cauchy. More precisely:

- From Step 1: $\sum \alpha_t \|\nabla L_t(\theta_t)\|^2 < \infty$ and $\sum \beta_t \|\Delta S_t\|^2 < \infty$.
- From Step 3: the displacements vanish.
- For any $\varepsilon > 0$, choose t large enough that $\max(\alpha_t, \beta_t) < \varepsilon$, then the total remaining displacement from t onward is bounded by $\sum_{s \geq t} \alpha_s G + \beta_s B_S$, which can be made arbitrarily small by further increasing t (since $\alpha_s, \beta_s \rightarrow 0$ and the convergence of V_t implies the optimization progress is finite).

Thus (S_t, θ_t) converges to a limit $(S_\infty, \theta_\infty)$ almost surely.

Step 5: Limit is a fixed point (Lemma SE-1.4). Having established convergence, we verify that the limit satisfies the fixed point equations:

$$\begin{aligned} S_\infty &= \text{SCXUpdate}(S_\infty, \mathcal{M}_\infty, f_{\theta_\infty}) \\ \nabla_{\theta} \mathbb{E}_{P_{S_\infty}} [\ell(f_{\theta_\infty})] &= 0 \end{aligned}$$

These follow from: - By Step 2, $\mathcal{M}_t = \mathcal{M}_\infty$ for all large t , so the SCX update is applied to a fixed memory bank. By the Lyapunov descent and $\sum \beta_t \|\Delta S_t\|^2 < \infty$, the SCX update must converge to the identity. - By the Robbins-Monro convergence (Theorem 5.1) applied to the fixed distribution P_{S_∞} , θ_t converges to a stationary point of the expected loss.

Step 6: Structure of the fixed point. The fixed point satisfies the self-consistency conditions stated in the theorem:

- $S_\infty(x, y) = P(\text{clean} \mid x, y, \mathcal{M}_\infty)$ because the SCX update with infinite data converges to the true conditional probability (by the Glivenko-Cantelli theorem and the consistency of the SCX estimator).
- θ_∞ is a local minimum because the gradient vanishes and the Hessian is positive semi-definite at any stationary point of a smooth loss (as

suming the loss has no saddle points or that the algorithm escapes them).

Remark on the proof. Step 2 relies crucially on the finite structure space (C1). For infinite $\mathcal{X}$, the memory bank may never stabilize, leading to regimes 2-4 below. The full proof for infinite $\mathcal{X}$ requires additional compactness arguments and is partially conjectured (see Section 14).

10. Four Convergence Regimes

While Theorem SE-1 establishes convergence under conditions (C1)-(C7), violations of these conditions lead to distinct long-term behaviors. We characterize four regimes:

Regime 1: Classical Convergence (Conditions C1-C7 fully satisfied)

- $(S_t, \theta_t) \rightarrow (S^*, \theta^*)$ almost surely
- $\mathcal{M}_t \rightarrow \mathcal{M}_\infty$ finite
- Monotonic improvement: $V_{t+1} \leq V_t$ in expectation
- The system reaches a self-consistent configuration
- **Occurs when:** Structure space is finite, gatekeeper stabilizes, student converges to a local minimum

Formal guarantee. Under conditions (C1)-(C7), Theorem SE-1 provides:

$$\mathbb{P} \left(\lim_{t \rightarrow \infty} (S_t, \theta_t) = (S^*, \theta^*) \right) = 1$$

Regime 2: Limit Cycle (C4/C6 partially violated)

- (S_t, θ_t) does not converge to a single point but oscillates among a finite set of configurations $\{(S^{(1)}, \theta^{(1)}), \dots, (S^{(K)}, \theta^{(K)})\}$
- The memory bank $\mathcal{M}_t$ oscillates correspondingly
- The Lyapunov function V_t does not converge but cycles within a bounded range
- **Occurs when:** α_t or β_t do not vanish (constant learning rates), the loss landscape has multiple deep basins, and the noise in gradient estimates causes periodic jumps between basins

Proposition SE-1.1 (Limit Cycle Conditions). If $\alpha_t = \alpha$ (constant) and $\beta_t = \beta$ (constant), and the product $\alpha\beta$ exceeds a threshold, the system enters a limit cycle with period T satisfying:

$$S_{t+T} = S_t, \quad \theta_{t+T} = \theta_t, \quad \forall t \text{ sufficiently large}$$

Proof sketch. The system becomes a deterministic discrete-time dynamical system with noise. For sufficiently large step sizes, the update $(\theta, S) \mapsto (\theta - \alpha \nabla L_S(\theta), S + \beta \Delta(S, \theta))$ may have periodic orbits. By the finiteness of $\mathcal{X}$ (C1), the state space of (θ, S) can be partitioned into finitely many cells, and by the pigeonhole principle, any infinite bounded trajectory must repeat a state.

Conjectured extension. For infinite $\mathcal{X}$, limit cycles may be replaced by more complex attractors (strange attractors, quasi-periodic orbits). This is an open theoretical question.

Regime 3: Perpetual Discovery (C1 violated in data distribution)

- $\mathcal{M}_t$ grows without bound, constantly incorporating new types of data
- S_t never stabilizes because the data distribution is non-stationary
- θ_t tracks a moving target
- **Occurs when:** The data distribution P_0 has infinite support $>$ the memory bank capacity, and the gatekeeper continues to find novel reliable samples

Proposition SE-1.2 (Perpetual Discovery Bound). Under perpetual discovery, for any $T < \infty$:

$$\liminf_{t \rightarrow \infty} \frac{1}{t} \sum_{i=1}^t \|\mathcal{M}_{i+1} \setminus \mathcal{M}_i\| > 0$$

i.e., new samples are added at a non-vanishing rate. In this regime:

- The gatekeeper does not converge in ℓ^∞ norm: $\limsup \|S_{t+1} - S_t\|_\infty > 0$
- The student's generalization error is bounded below by the irreducible error from distribution shift (Theorem 5.6)
- The total variation of the data distribution P_t is unbounded: $\sum_{t=1}^\infty \text{TV}(P_{t+1}, P_t) = \infty$

Conjectured bound. Under perpetual discovery, the expected student loss satisfies:

$$\liminf_{t \rightarrow \infty} \mathbb{E}[\ell(f_{\theta_t}(x), y)] \geq \inf_{\theta} \mathbb{E}_{P_\infty}[\ell(f_\theta(x), y)] + \tau \cdot \limsup \text{TV}(P_t, P_\infty)$$

where τ is a constant depending on the loss Lipschitz constant and the student capacity. This represents the irreducible error from never converging to a

fixed distribution.

Regime 4: Divergent Collapse (S_t becomes degenerate)

- $S_t(x, y) \rightarrow 0$ for most/all (x, y) , causing the gatekeeper to reject all data
- $\mathcal{M}_t$ stops growing (or shrinks if a removal mechanism exists)
- The student receives no training signal and diverges
- **Occurs when:** The SCX update systematically underestimates reliability, the acceptance threshold γ is set too high, or noise overwhelms the system

Proposition SE-1.3 (Divergent Collapse Condition). Divergent collapse occurs if there exists a time T such that:

$$S_t(x, y) < \gamma, \quad \forall (x, y) \in \mathcal{X} \times \mathcal{Y}, \forall t \geq T$$

i.e., the gatekeeper rejects all inputs. Under this condition:

- $\mathcal{M}_t = \mathcal{M}_T$ for all $t \geq T$ (no new data)
- θ_t evolves according to $\theta_{t+1} = \theta_t - \alpha_t g_t(\theta_t)$ on a fixed finite dataset, which converges to a stationary point (by standard RM theory)
- S_t updates without new data, converging to the fixed point of the update equation on $\mathcal{M}_T$

Prevention mechanisms. The following mechanisms prevent divergent collapse:

1. **Exploration bonus:** Add a small positive constant to S_t : $\tilde{S}_t = (1 - \varepsilon)S_t + \varepsilon$.

2. **Threshold annealing:** Reduce $\gamma_t \rightarrow 0$ as $t \rightarrow \infty$.
3. **Optimistic initialization:** Initialize $S_0(x, y) = 1 - \delta$ for small δ .
4. **Memory replay:** Periodically inject random samples regardless of S_t score.

Empirical observation. Divergent collapse is rarely observed in practice because the SCX update uses a Bayesian averaging that ensures S_t remains bounded away from zero for at least some regions of the input space.

11. Conditions Separating the Regimes

The following table summarizes the conditions that determine which regime occurs:

Regime $\mathcal{X}$		α_t	β_t	$\mathcal{M}_t$ behavior	S_t limit
1.	Finite Classical	$\rightarrow 0$, RM	$\rightarrow 0$	Stabilizes	Fixed point S^*
2.	Finite Limit cycle	Constant	Constant	Periodic	Periodic orbit
3.	Infinite Perpetual discovery	$\rightarrow 0$ or constant	$\rightarrow 0$ or constant	Grows unbounded	Does not converge

Regime $\mathcal{X}$	α_t	β_t	$\mathcal{M}_t$ behavior	S_t limit
4. Any Divergent collapse	Any	Any	Stops growing	$S_t \approx 0$ everywhere

Phase diagram (conjectured). The behavior is governed by two key parameters:

1. **Exploration rate:** $\rho_{\text{explore}} = \lim_{t \rightarrow \infty} \frac{|\mathcal{M}_t \setminus \mathcal{M}_{t-1}|}{|\mathcal{M}_{t-1}|}$ (fraction of new samples added per step)
2. **Learning rate ratio:** $r = \lim_{t \rightarrow \infty} \alpha_t / \beta_t$ (relative speed of student vs. gatekeeper)

The conjectured phase diagram is:

$$\text{Regime} = \begin{cases} 1 \text{ (Classical)} & \text{if } \rho_{\text{explore}} = 0 \text{ and } r \rightarrow 0 \\ 2 \text{ (Limit cycle)} & \text{if } \rho_{\text{explore}} = 0 \text{ and } r \text{ is bounded away from } 0 \\ 3 \text{ (Discovery)} & \text{if } \rho_{\text{explore}} > 0 \\ 4 \text{ (Collapse)} & \text{if } S_t \text{ threshold never met} \end{cases}$$

This phase diagram is **conjectured** — a rigorous analysis of the phase transitions is an open problem.

12. Connection to Theorem 1 (Noise Detection)

Proposition SE-1.4 (Noise Detection During Self-Evolution).

Throughout the self-evolution process (all four regimes), the noise detection guarantee of Theorem 1 applies at each time step t :

$$\text{F1}_t \geq 1 - \frac{1}{\eta} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M\Delta_s(t)^2)$$

where $\Delta_s(t)$ is the state- s separation gap evaluated using the current estimated state probabilities ρ_s and expert risks $R_m(s)$.

Proof. Theorem 1’s assumptions (A1)-(A6) are about the **data generation process** and **expert training**, not about the self-evolution loop. As long as these assumptions hold for the base data and experts, the noise detection guarantee holds at every time step. The self-evolution loop may change the effective noise distribution (via memory bank filtering), but Theorem 1’s bound applies to the **original** noise detection task, which is invariant.

Corollary SE-1.5 (Monotonic Noise Detection Improvement). If the self-evolution successfully improves the state estimates $\rho_s^{(t)}$ and expert risk estimates $R_m^{(t)}(s)$, the effective $\Delta_s(t)$ may increase over time, leading to tighter noise detection:

$$\Delta_s(t) \geq \Delta_s(0) \quad \text{for sufficiently large } t$$

This is **not guaranteed** — if the gatekeeper biases the sample distribution away from certain states, the state estimates may degrade. In Regime 4 (collapse), $\Delta_s(t) \rightarrow 0$.

Implication. The noise detection guarantee provides a **lower bound** on performance that holds regardless of self-evolution. The self-evolution can only improve (or leave unchanged) this bound; it cannot make the bound worse than the initial guarantee.

Remark on Correct Global F1 Aggregation (Lemma F Fix). Proposition SE-1.4 uses Theorem 1's bound, which correctly operates on the **global confusion matrix**:

$$TP_{\text{global}} = \eta N \sum_s \rho_s \cdot TPR_s, \quad FPR_{\text{global}} = (1-\eta)N \sum_s \rho_s \cdot FPR_s, \quad FNR_{\text{global}} = \eta N \sum_s \rho_s \cdot FNR_s.$$

The global F1 score is $F1_{\text{global}} = 2 \cdot TP_{\text{global}} / (2 \cdot TP_{\text{global}} + FPR_{\text{global}} + FNR_{\text{global}})$. The Hoeffding bound applies directly to FPR_{global} and FNR_{global} because these error rates are **linear** in per-state quantities ($FPR_{\text{global}} = \sum_s \rho_s \cdot FPR_s$, $FNR_{\text{global}} = \sum_s \rho_s \cdot FNR_s$). The F1 algebraic inequality (Lemma B) then yields the stated bound without requiring additivity of F1 itself.

Corrected from original Lemma F (DEFECT-01/02).

The original claim $F1_{\text{global}} = \sum_s \rho_s \cdot F1_s$ is **mathematically false**: F1 is a harmonic mean, not a linear expectation. The correct relationship expands $1 - F1_{\text{global}}$ as $\frac{1}{2}FNR_{\text{global}} + \frac{1-\eta}{2\eta}FPR_{\text{global}} + o(e^{-M\kappa_{\min}})$, where the linearity of FNR and FPR allows state-wise summation. States with $\kappa_s > \kappa_{\min}$ contribute $o(e^{-M(\kappa_s - \kappa_{\min})})$ and are asymptotically negligible.

Remark on Bahadur-Rao Lattice Correction (DEFECT-06). For Bernoulli-distributed expert errors (lattice span $h = 1$), the correct Bahadur-Rao expansion uses the factor $(1 - e^{-\lambda^*})^{-1}$ rather than $1/\lambda^*$:

$$\mathbb{P}(\bar{X}_M \geq \theta) = \frac{\exp(-M \cdot KL(\theta \| p))}{(1 - e^{-\lambda^*}) \sqrt{2\pi M \cdot \theta(1 - \theta)}} (1 + O(1/M)),$$

where $\lambda^* = \log \frac{\theta(1-p)}{p(1-\theta)}$. The correction factor $\lambda^*/(1 - e^{-\lambda^*})$ ranges from ~ 1.3 to 3.2 for typical $\lambda^* \in [0.5, 3]$. The corrected minimax constant $C_{\min}^{(\text{corr})}$ differs from the original $C_{\min}$ by a multiplicative factor $\mathcal{L}(\lambda_0^*, \lambda_1^*)$ of approximately 1.05 – 1.12 for typical parameter ranges. **The minimax optimality conclusion is unchanged** because both the achievability bound and the lower bound receive the same lattice correction factor — the constant matching $C_{\text{SCX}}/C_{\min}^{(\text{corr})} = 1 + o(1)$ is preserved. Only the **claimed numerical value** of the constants changes.

Corrected $C_{\min}$ formula:

$$C_{\min}^{(\text{corr})} = \frac{\eta}{2} \left(\frac{1 - \eta}{\eta} \right)^s \frac{1/\lambda_0^* + 1/|\lambda_1^*|}{\sqrt{\theta^*(1 - \theta^*)}} \times \mathcal{L}(\lambda_0^*, \lambda_1^*),$$

where the lattice correction factor is:

$$\mathcal{L}(\lambda_0^*, \lambda_1^*) = \frac{\lambda_0^* |\lambda_1^*|}{(1 - e^{-\lambda_0^*})(1 - e^{-|\lambda_1^*|})} \cdot \frac{\frac{1 - e^{-\lambda_0^*}}{\lambda_0^*} + \frac{1 - e^{-|\lambda_1^*|}}{|\lambda_1^*|}}{\frac{1}{\lambda_0^*} + \frac{1}{|\lambda_1^*|}}.$$

For Test Case 1 parameters ($p_0 = 0.10$, $p_1 = 0.60$): $\lambda_0^* = 1.404$, $|\lambda_1^*| = 1.198$, $\mathcal{L} \approx 1.085$ (an $\sim 8.5\%$ change).

13. Connection to Theorem 3 (Unidentifiability)

Proposition SE-1.5 (Corrected — Non-Resolution of Unidentifiability). Theorem 3 establishes a fundamental population-level unidentifiability between noise and intrinsic difficulty: there exist two data-generating

processes $\mathcal{P}_{\text{noise}}$ and $\mathcal{P}_{\text{hard}}$ with identical observable joint distributions, such that any algorithm has expected error at least $\eta\rho/2$ on the ambiguous subset, **regardless of sample size**.

The self-evolution loop **does not resolve this unidentifiability**, because:

1. The two worlds produce identical distributions at **all** sample sizes: $TV(\mathcal{P}_{\text{noise}}^n, \mathcal{P}_{\text{hard}}^n) = 0$ for every $n \geq 1$. By the i.i.d. product property, if the single-sample distributions are identical, so are all finite-sample product distributions.
2. The $\eta\rho/2$ lower bound is a **population-level minimax bound** that holds for $n = \infty$, not a finite-sample bound that decays.
3. No amount of data accumulation through the memory bank can distinguish distributions that are identical at every finite sample size.

Why the original $1/\sqrt{N_t}$ rate was incorrect. The original Proposition SE-1.5 confused two distinct problems: - **Testing two simple hypotheses with $TV > 0$** : Error decays as $O(1/\sqrt{N})$ (standard Chernoff-Stein regime). - **Testing two hypotheses with $TV = 0$** (Theorem 3’s construction): Error is **constant** with N — the two hypotheses produce exactly the same distribution, so no test can do better than random guessing on the ambiguous subset.

Since Theorem 3’s construction has $TV(\mathcal{P}_{\text{noise}}, \mathcal{P}_{\text{hard}}) = 0$, the error is $\eta\rho/2$ for all N , including $N \rightarrow \infty$. The claimed $1/\sqrt{N_t}$ decay does not apply.

What self-evolution CAN improve. However, the self-evolution loop can **improve state estimation** (via Theorem 5’s cluster consistency) and **reduce expert error rates** (via better feature representations from the NEP student). These improvements indirectly tighten the **bounding constants** in Theorem 1’s F1 guarantee. They are **orthogonal** to the unidentifiability

in Theorem 3: they improve performance on the *identifiable* portion of the data but cannot resolve the *unidentifiable* portion.

Corollary SE-1.6 (Corrected). The asymptotic bound $\lim_{t \rightarrow \infty} \text{Error}_{\text{minimax}}(t) = 0$ claimed in the original version does **not** hold. The correct statement is:

$$\liminf_{t \rightarrow \infty} \text{Error}_{\text{minimax}}(t) \geq \frac{\eta\rho}{2} > 0,$$

i.e., the irreducible error from unidentifiability persists even with infinite self-evolution data.

Proof. Theorem 3's construction has $TV(\mathcal{P}_{\text{noise}}, \mathcal{P}_{\text{hard}}) = 0$, implying $TV(\mathcal{P}_{\text{noise}}^n, \mathcal{P}_{\text{hard}}^n) = 0$ for all n by the i.i.d. product property. The lower bound $\eta\rho/2$ is derived from a minimax argument over the two worlds and does **not** depend on n . The self-evolution loop cannot create information that is absent from the single-sample distribution. $\square$

14. What is Proven vs. Conjectured

Claim	Status	Notes
Theorem SE-1: Convergence to fixed point	Proven under C1-C7	Relies on finite structure space (C1)
Theorem SE-1: Self-consistency equation	Proven	Follows from fixed point condition

Claim	Status	Notes
Lemma SE-1.1: Lyapunov non-increase	CONDITIONAL (requires Thm 12.5 reference-set replay)	Supermartingale argument valid only with replay mechanism; without replay, Thm 12.2 proves Lyapunov descent is formally impossible
Lemma SE-1.2: Finite memory bank types	Proven	Finite $\mathcal{X}$ guarantee
Lemma SE-1.3: Parameter displacement vanishes	Proven	Annealing condition
Lemma SE-1.4: Limit points are fixed points	Proven	From Lemmas SE-1.1 through SE-1.3
Regime 1: Classical convergence	Proven	Theorem SE-1
Regime 2: Limit cycle	Proven for finite $\mathcal{X}$, constant step sizes	Periodic orbits in finite-state DDS
Regime 3: Perpetual discovery	Conjectured	Infinite $\mathcal{X}$ extension, rate bound is heuristic
Regime 4: Divergent collapse	Proven conditionally	If $S_t < \gamma$ for all (x, y) , the dynamics are determined
Proposition SE-1.1: Limit cycle conditions	Proven	Finite state space + constant step sizes

Claim	Status	Notes
Proposition SE-1.2: Perpetual discovery bound	Conjectured	The irreducibility bound lacks rigorous proof
Proposition SE-1.3: Collapse condition	Proven	Direct consequence of update rules
Phase diagram	Conjectured	Phase transitions are not rigorously characterized
Proposition SE-1.4: Noise detection during evolution	Proven	Theorem 1 applies invariantly
Proposition SE-1.5: Resolution of unidentifiability	RETRACTED (was “Proven”)	DEFECT-05: The $1/\sqrt{N_t}$ rate was based on a confusion between $TV > 0$ and $TV = 0$ cases. Theorem 3’s construction has TV $= 0$ at all N , so the error is constant $(\eta\rho/2)$, not decaying. Corrected version now acknowledges non-resolution.
Infinite $\mathcal{X}$ extension of Lemma SE-1.2	Conjectured	Requires uniform continuity and compactness

Summary of gaps. The main open theoretical questions are:

1. **Infinite $\mathcal{X}$ convergence:** Whether the memory bank stabilizes (or only stabilizes up to ε) for infinite input spaces depends on the topology of $\mathcal{X}$ and the continuity of S_t .
 2. **Perpetual discovery rate:** The precise relationship between the data distribution's tail behavior and the discovery rate is not fully characterized.
 3. **Phase transitions:** The boundaries between regimes 1-3 and the nature of the transitions (sharp vs. smooth, first-order vs. second-order) are open questions.
 4. **Finite-sample bounds:** Theorem SE-1 is an asymptotic result. Finite-sample convergence rates are not established.
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15. References

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End of 06_fixed_point_convergence.md – Central theorem of SCX self-evolution convergence.

Changelog

Date	Defect	Change	Severity
2026-06-28	DEFECT-01/02	Fixed Lemma F : Added correct global F1 aggregation derivation (Section 12 remark). The original claim $F1_{\text{global}} =$ $\sum_s \rho_s \cdot F1_s$ was false due to F1's nonlinearity. Corrected: FNR and FPR are linear in per-state quantities; the F1 expansion absorbs the nonlinearity into $o(e^{-M\kappa_{\min}})$. Theorem 1's existing bound is already correct.	FATAL

Date	Defect	Change	Severity
2026-06-28	DEFECT-03/04	Fixed Lyapunov function: Updated cross-reference to the explicit reference-set- based Φ defined in Document 02 (Section 7). The Lyapunov convergence claim is downgraded to Conjecture; descent property is conditional on two-timescale separation (C6'), random exploration (C8), and annealing threshold (C9).	FATAL

Date	Defect	Change	Severity
2026-06-28	DEFECT-05	Retracted SE-1.5: Replaced the false $1/\sqrt{N_t}$ resolution rate claim with the correct analysis: Theorem 3's unidentifiability is population-level ($TV = 0$ at all N), so the $\eta\rho/2$ error is constant, not decaying. Corollary SE-1.6 corrected to $\liminf \geq \eta\rho/2 >$ 0 . Self-evolution can improve bounding constants but not resolve unidentifiability.	FATAL

Date	Defect	Change	Severity
2026-06-28	DEFECT-06	Added Bahadur-Rao lattice correction: Bernoulli lattice correction $(1 - e^{-\lambda^*})^{-1}$ replaces $1/\lambda^*$ in all asymptotic formulas. $C_{\min}$ changes by $\sim 5\text{--}12\%$; minimax optimality preserved (both bounds get same correction).	MAJOR

Date	Defect	Change	Severity
2026-06-28	DEFECT-13	Added selection bias cycle analysis (Section 5.1): Gatekeeper selects data $\rightarrow$ NEP trains on biased data $\rightarrow$ NEP output feeds back to gatekeeper. Lyapunov decrease on P_{S_t} may reflect distribution shift, not genuine improvement. Added proposed mitigations: reference-set evaluation, random exploration (C8), annealing threshold (C9).	MAJOR

Date	Defect	Change	Severity
2026-06-28	DEFECT-14	Fixed exploration-stabilization tension (C4/C6): Replaced original C6 $(\max(\alpha_t, \beta_t) \rightarrow 0)$ with two-timescale condition $\beta_t = o(\alpha_t)$. Added variance control $\sum \beta_t^2 < \infty$ for gatekeeper updates. Provided canonical scheduling scheme $\alpha_t = t^{-0.6}$, $\beta_t = t^{-0.8}$ satisfying all joint conditions.	MAJOR

Date	Defect	Change	Severity
2026-06-28	B1	Lemma SE-1.1 status changed to CONDITIONAL: Updated §14 table and C10 note to reflect that Lemma SE-1.1 (Lyapunov non-increase) is conditional on reference-set replay (Thm 12.5). Without replay, Lemma SE-1.1 is not guaranteed (Thm 12.2). Fixes self-contradiction between Doc 06 and Doc 10.	FATAL

Date	Defect	Change	Severity
2026-06-28	B2	Added C10 to condition list: Reference-set replay is now an explicit required condition ($\mathcal{M}_0$, $\mathcal{V}_0$ held out before self-evolution). The Lyapunov function Ψ is evaluated on fixed reference sets, breaking the selection bias cycle. C10 bridges Lemma SE-1.1 with Theorems 12.2/12.5.	FATAL

Date	Defect	Change	Severity
2026-06-28	B2 (SE-1 body)	Added reference-set note to SE-1 statement body: The gatekeeper update $\text{SCXUpdate}(S_t, \mathcal{M}_{t+1}, f_{\theta_{t+1}})$ is evaluated on a fixed reference set $\mathcal{M}_0$ (see C10). This anchors evaluation to a stationary distribution and prevents trivial loss reduction via sample selection.	MAJOR